AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH
Faculty of Science and Technology
Department of Computer Science
CSC 3215 Web Technologies

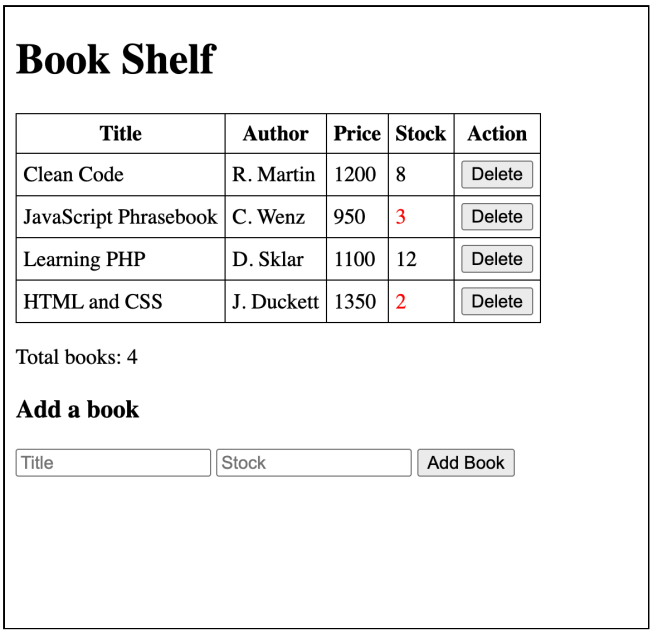
Practice Task: Book Shelf (JavaScript DOM)

Course: CSC 3215 Web Technologies | **Time:** 20 minutes | **Type:** In-class practice (not graded)

Create **one HTML file** named `books.html` with internal CSS and internal JavaScript (a `<style>` tag and a `<script>` tag in the same file). Build the page using JavaScript DOM methods. Do not type the table rows in HTML.

Expected result

Your finished page should look like this (the result after Parts 1 to 4, before any book is added):



Data

Copy this array into your script:

```
let books = [
    { title: "Clean Code", author: "R. Martin", price: 1200, stock: 8 },
    { title: "JavaScript Phrasebook", author: "C. Wenz", price: 950, stock: 3 },
    { title: "Learning PHP", author: "D. Sklar", price: 1100, stock: 12 },
    { title: "HTML and CSS", author: "J. Duckett", price: 1350, stock: 2 }
];
```

Page layout (HTML)

- An `<h1>` with the text “Book Shelf”.
- An empty `<div>` with the id `shelf`, where the table will be placed.
- A `<p>` with the id `count`.
- Two inputs (a text input for the title and a number input for the stock), a button “Add Book”, and a `<p>` with the id `msg`.

- Give the table a 1px solid black border with collapsed borders and 5px padding in the cells (use CSS in the `<style>` tag).

Part 1: Build the table (about 7 minutes)

1. Create the table with JavaScript and place it inside the `shelf` div.
2. Add a header row with the headings **Title, Author, Price, Stock, Action**.
3. Write a function `addBookRow(book)` that creates one table row for a book. Call it for every book in the array using `forEach`.
4. Show the text "Total books: 4" inside the `count` paragraph, using the length of the array.

Part 2: Highlight low stock (about 2 minutes)

When a book row is created, if the stock is **less than 5**, show the stock number in **red** using `element.style`.

Part 3: Add a book (about 6 minutes)

When the "Add Book" button is clicked:

1. Read the title and the stock from the two inputs.
2. If the title is empty, show "Title is required" in red inside the `msg` paragraph and do not add anything.
3. Otherwise, create a book object with the entered title and stock, the author "Unknown" and the price 0. Add it to the `books` array, add its row to the table by calling `addBookRow`, and update the total in the `count` paragraph.
4. Clear the two inputs after a successful add.

Part 4: Delete a book (about 5 minutes)

1. Add a **Delete** button in the last cell of every row, including the rows of books added in Part 3.
2. When a Delete button is clicked, remove that row from the table, remove that book from the `books` array, and update the total in the `count` paragraph.

Hint: this was not shown in the slides. Search the web for "how to remove an HTML element using JavaScript" and "how to remove an element from an array by index" (`indexOf()` is already in your slides). Pay attention to which element the button belongs to.

Check your work

- The page shows 4 books, and the stock values 3 and 2 are red.
- Adding "Test Book" with stock 3 adds a fifth row with a red stock value, and the total shows 5.
- Clicking "Add Book" with an empty title adds nothing and shows a red message.
- Clicking Delete on any row removes that row and lowers the total by 1.

Show the finished page to your instructor.